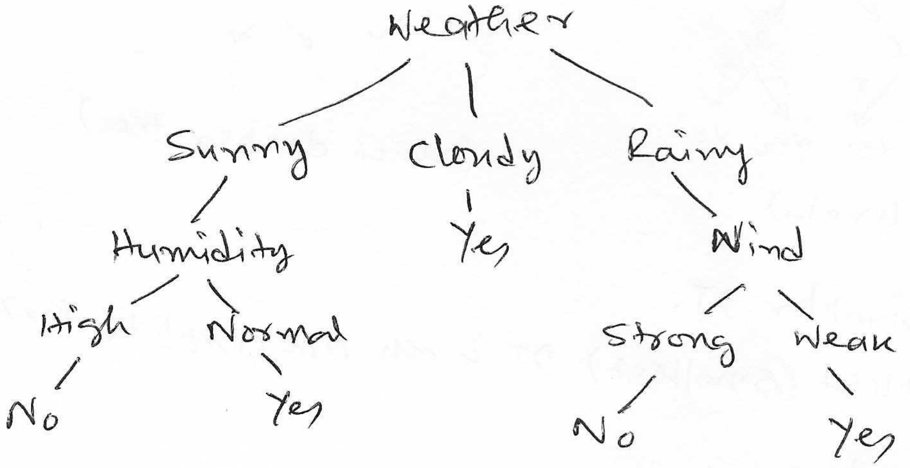
Decision Trees

Decision Tree analysis is in general a predictive modeling tool that has applications on number of areas. It's a tree-like graph with nodes representing the place we pick an attribute, & ask a question; edges represent the answers to the question; and leaves represent the actual output or class label.

Ex: We want to play cricket on a particular day - say Saturday. How will you decide whether to play or not? Let's say you go out & check different weather conditions. You take all these factors into account to decide if you want to play or not. So you calculate all these factors for last 10 days & form a lookup table.

Day	Weather	Temperature	Humidity	Wind	Play?
1.	Sunny	Hot	High	Weak	No
2.	Cloudy	Hot	High	Weak	Yes
3.	Sunny	~~Normal~~ Mild	~~Strong~~ Normal	Strong	Yes

⋮



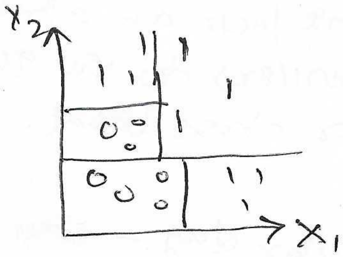
DT learning is a method for approximating discrete valued target function.

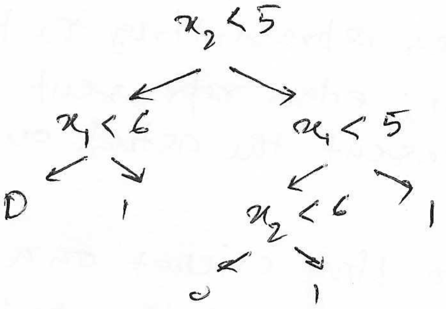
General algo

1. Pick the best attribute / feature.
2. Ask the relevant question.
3. Follow the answer path.
4. Go to step-1 until you arrive to the answer.

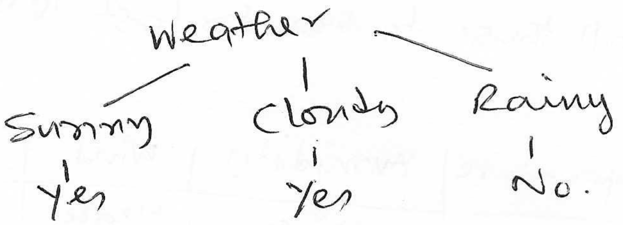
Decision boundaries

DTs divide the feature space into axis parallel rectangles, & label each rectangle with one of the k classes.





Decision Stump $\Rightarrow$ One level decision tree.

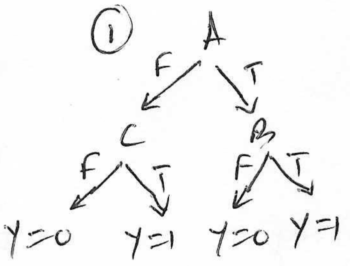


Not all DTs will have the same size.

E.g. $Y = (A \land B) \lor (\neg A \land C)$

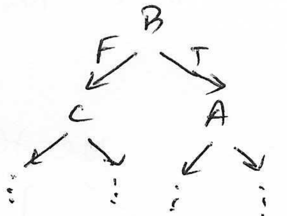
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

1



(2-levels)

2



(much deeper tree)

Goal! Learn a simpler DT.

[Learning the simplest (smallest) DT is an NP-Complete problem].

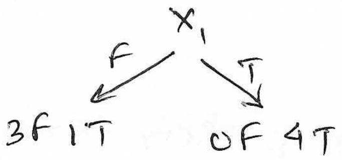
Two important questions

- Which features to choose
- When do you stop.

Choosing a good feature to branch.

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F
F	T	F
F	F	F

Before split: $\frac{3}{8}$ false, $\frac{5}{8}$ true.

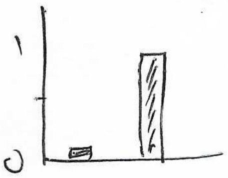


So the dataset becomes "simpler" on both sides (like majority vote).

So we need to know some quantitative measure so that we know whether splitting on X_1 or X_2 is better.

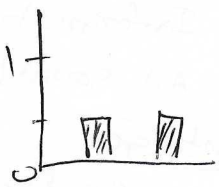
⇓

Entropy!! → "good" vs "bad" distribution.

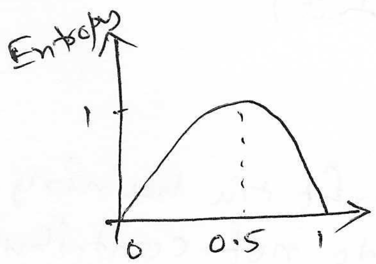


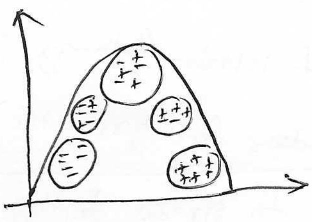
Good one

Bad one



More uncertainty → more entropy.





$$H(Y) = -\sum_{i=1}^{k} P(Y=y_i)\log_2 P(Y=y_i)$$

Mutual Information Gain

$$I(Y, X) = H(Y) - H(Y|X)$$

How much uncertainty is reduced

How uncertain we are initially

How uncertain we are after seeing X.

If X is useless, $H(Y|X) \approx H(Y)$

⇓

$I(Y, X) \approx 0$ → no information gain → bad feature.

$$x^* = \arg\max_x I(Y, X)$$

Since $H(Y)$ is constant w.r.t. X,

$$x^* = \arg\min H(Y|X)$$

Pruning

A challenge with DT is their tendency to overfit the training data, especially when they're allowed to grow without constraints. To address this, pre-pruning & post-pruning are used.

Pre-pruning (early stopping)

- Tree reaches a predefined max^m depth.
- No. of samples is below threshold.
- Information gain is less than a threshold.
- All samples in the node belong to same class.

Advantages

- Reduces overfitting
- Faster training.

Disadvantages

- Risk of underfitting (too shallow tree)

Post-pruning (after full growth)

Allow the DT to grow fully, allow to overfit the training data, then trim back unnecessary branches that do not contribute to improving accuracy on unseen data (validation set).

Advantages

- Better generalization
- Avoids overfitting.

Disadvantages

- Slower (needs to grow the fully)
- Validation set required.

Grow during training, prune during validation.

Gini impurity (GI)

Measures how mixed (or impure) the classes are in a node.

⇒ Node contains one class → pure (GI = 0)

⇒ Node contains mix of classes → impure (GI > 0)

Low GI is always preferred.

In other words, GI measures the probability of incorrectly classifying a randomly chosen element from the dataset if it were labelled according to the class distribution in that set.

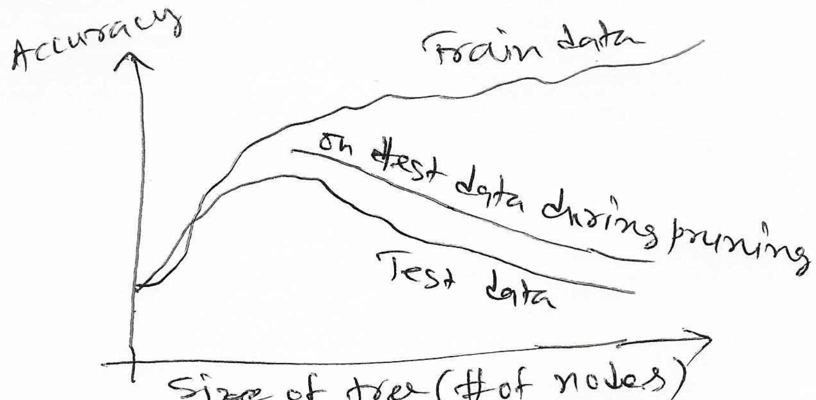
$$\text{Gini}(S) = 1 - \sum_{i=1}^{k} p_i^2,$$

where p_i is the proportion of class c_i in the dataset.

Entropy vs. GI

- Both measures impurity, but GI is usually faster.
- Entropy tends to penalize impurity a bit more heavily.
- In practice, both give similar results building DTs.

Effect of pruning in general.



Usually pruning gives better outcome.